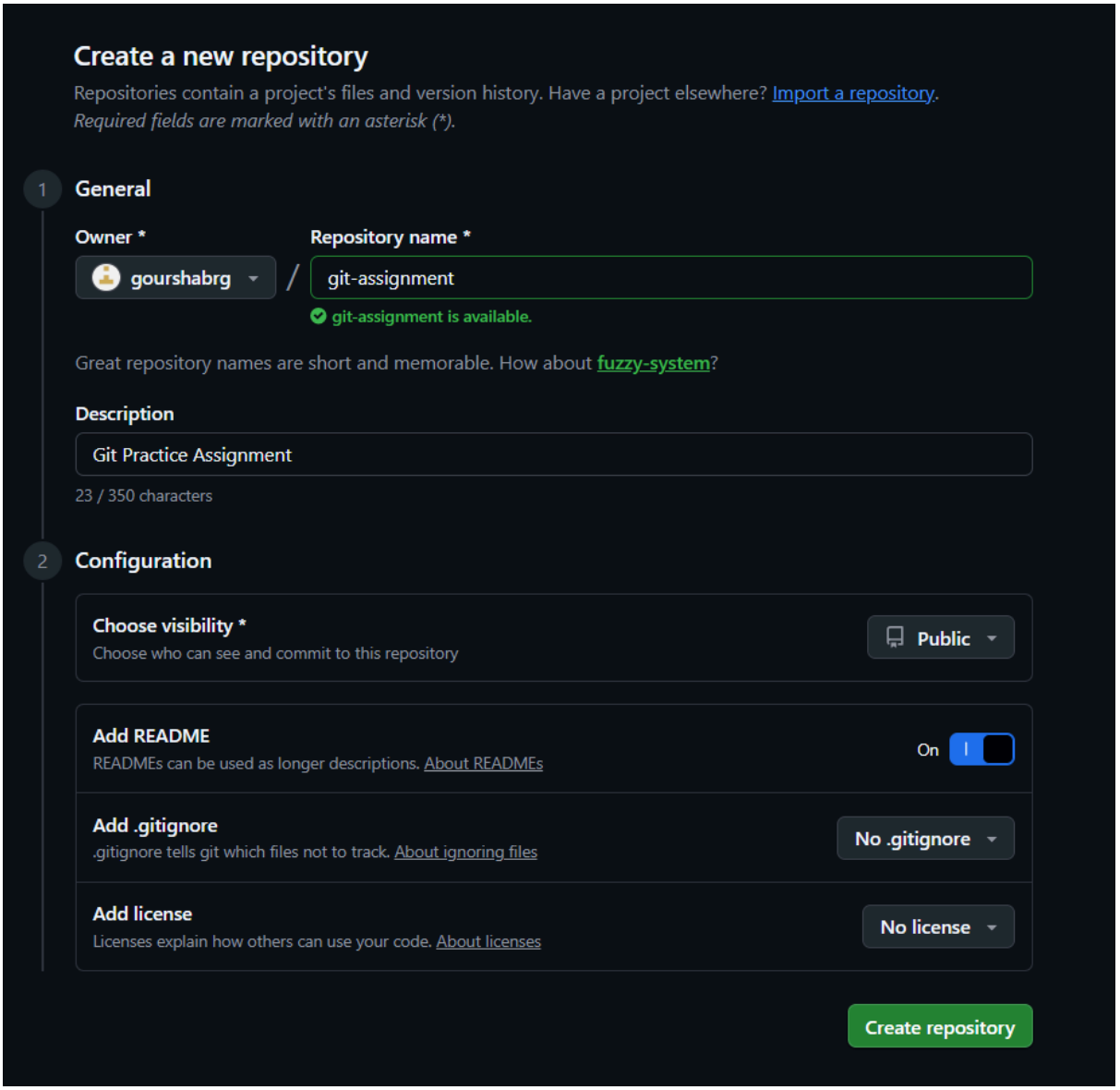
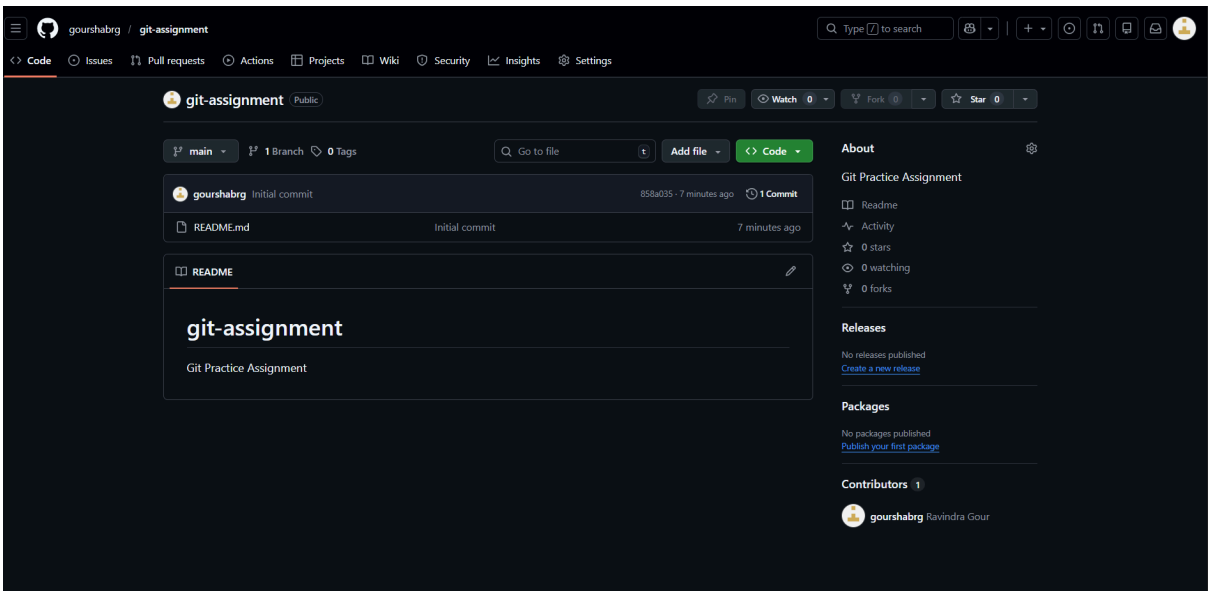
Submitted by: Ravindra Gour
Repository:https://github.com/gourshabrg/Assignment

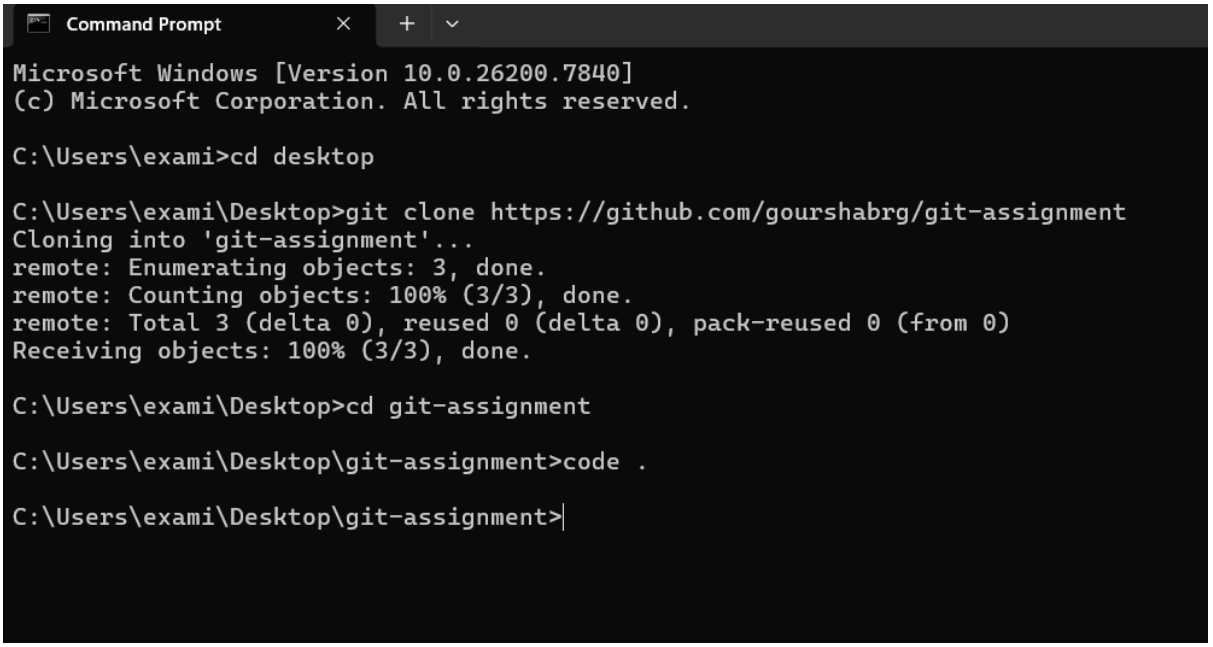
1. **Create a New Repository**
    1. Go to https://github.com
    2. Click **New Repository**
    3. Enter:
        - Repository name
        - Description
        - Add Readme
    4. Select **Public**
    5. **Click Create repository**





2. Clone Repository to Your System

1. Open **Git Bash / Terminal** and enter this command (**git clone _repo_url).**
    - **git clone** https://github.com/gourshabrg/Assignment
2. Then go inside the folder using **cd git-assignment (cd _folder_name).**
3. Also open in any code editor i used **Vscode .**

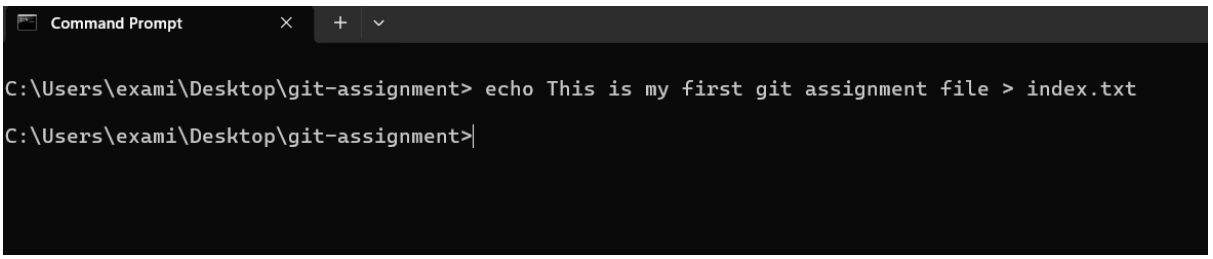


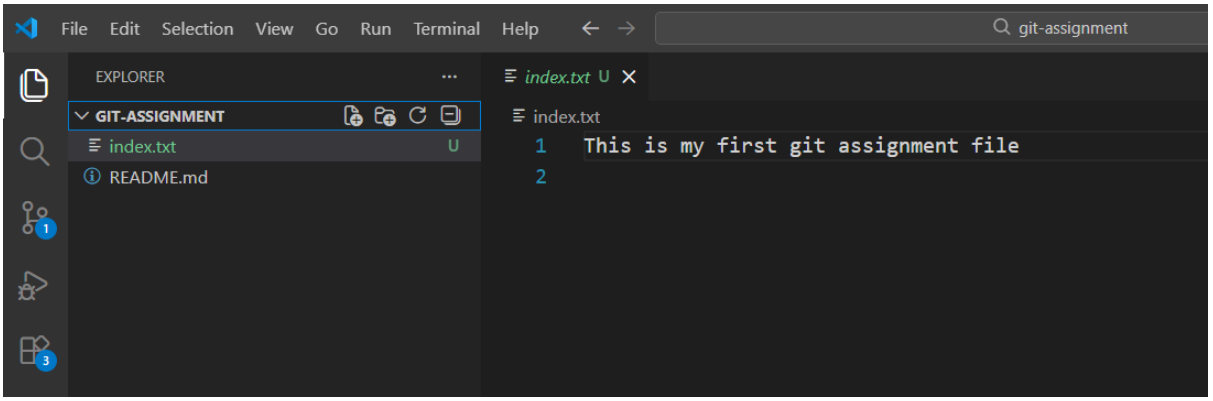
3. Create a New File

Using **Cmd** or also use the **Code editor**

echo [content] > [file_name]

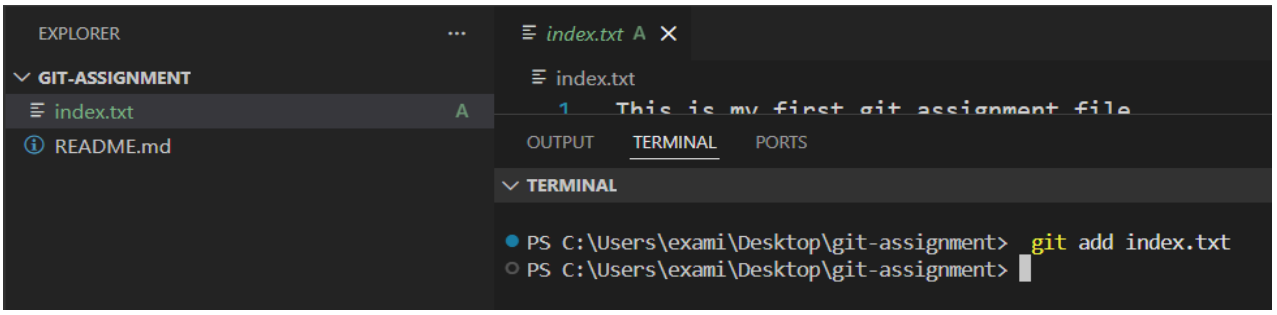
Learn about : echo



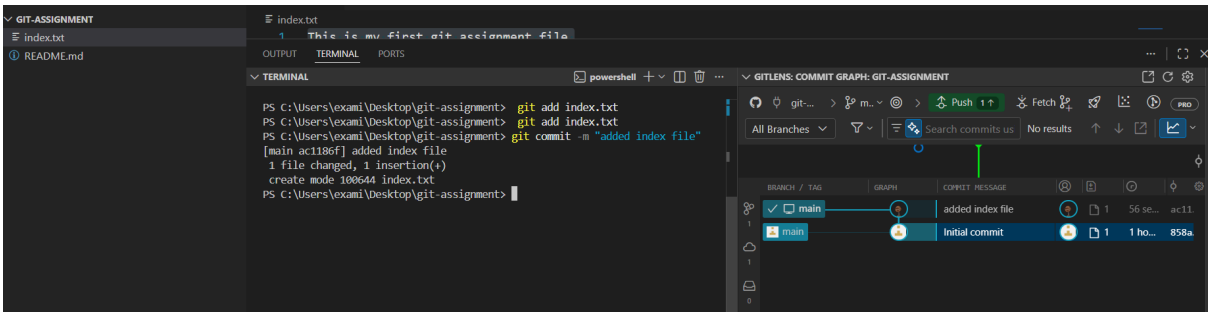


Make the file trackable in Git

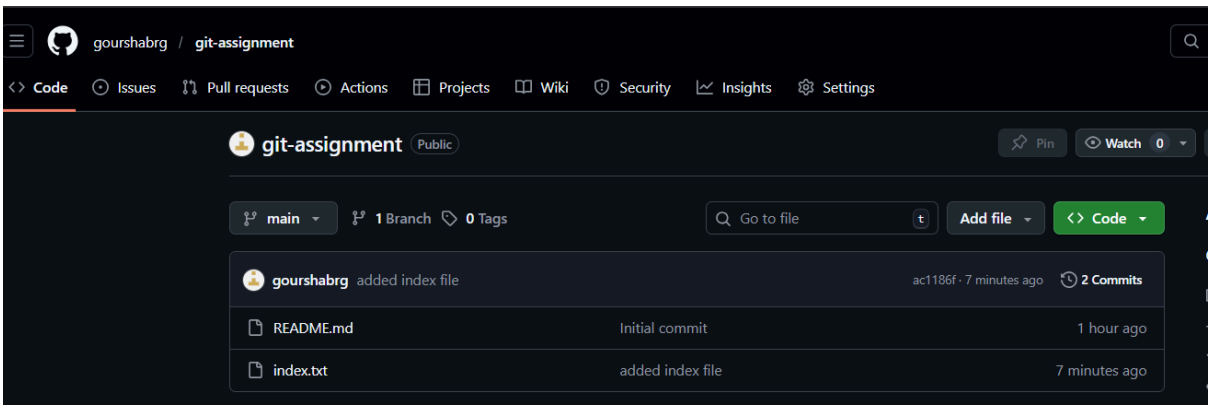
1. **git add [File_Name]**
   About: This command adds the file to the staging area.



2. **git commit** -m "Added index file"
   About: This command saves the staged changes permanently in the local Git repository.



3. **git push origin main**
   About: This command uploads your committed changes to the remote repository (like GitHub).



4. Create a Branch

- **git branch [new_branch_name]**
  About: used to **create a new branch in Git.**
- **git checkout [new_branch_name]**
  About: This command **switches your working directory to the new branch.**
- **git checkout -b feature1**
  About: This command **creates a new branch and switches to it immediately.**

```
PS C:\Users\exami\Desktop\git-assignment> git branch feature1
PS C:\Users\exami\Desktop\git-assignment> git checkout feature1
Switched to branch 'feature1'
PS C:\Users\exami\Desktop\git-assignment> git loh
git: 'loh' is not a git command. See 'git --help'.

The most similar command is
        log
PS C:\Users\exami\Desktop\git-assignment> git log
commit ac1186fdb0f3adec3709f2dbfec89d7193eb086d (HEAD -> feature1, origin/main, origin/HEAD, main)
Author: Gourshab <ravindragourcode786@gmail.com>
Date:   Wed Mar 11 10:27:02 2026 +0530

    added index file

commit 858a0353cd99f903b2e8b7e9d9f263e119538911
Author: Ravindra Gour <143959353+gourshabrg@users.noreply.github.com>
Date:   Wed Mar 11 09:26:33 2026 +0530

    Initial commit
PS C:\Users\exami\Desktop\git-assignment>
```

5. Make Changes in the New Branch

- Edit the file:

Use command

- **git add .**
- **git commit -m "message"**

About: This command saves the staged changes as a new commit in the local Git repository with a message describing the changes.

- **git push origin feature1**

About: This command uploads the local branch feature1 to the remote repository (origin) such as GitHub.

6. Create Another Branch

- First go to the main branch using **git checkout main.**

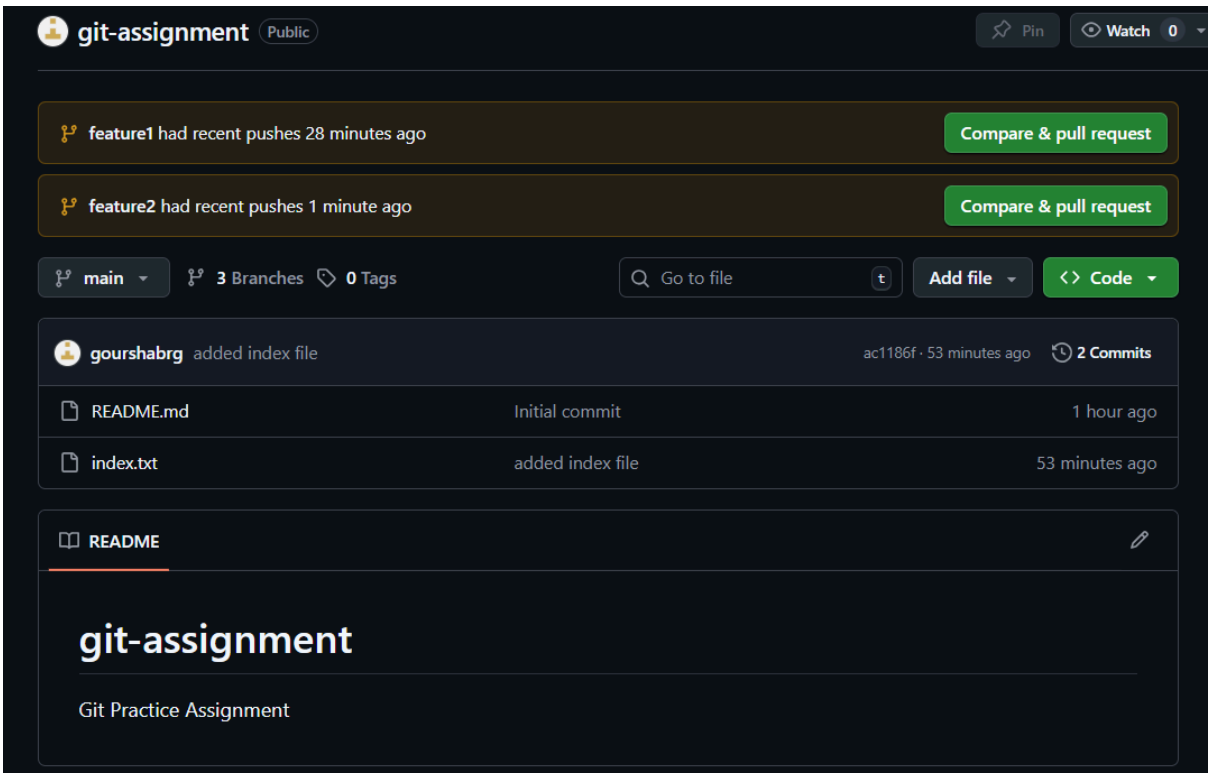
```
PS C:\Users\exami\Desktop\git-assignment> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\exami\Desktop\git-assignment> git log
commit ac1186fdb0f3adec3709f2dbfec89d7193eb086d (HEAD -> main, origin/main, origin/HEAD)
Author: Gourshab <ravindragourcode786@gmail.com>
Date:   Wed Mar 11 10:27:02 2026 +0530

    added index file

commit 858a0353cd99f903b2e8b7e9d9f263e119538911
Author: Ravindra Gour <143959353+gourshabrg@users.noreply.github.com>
Date:   Wed Mar 11 09:26:33 2026 +0530

    Initial commit
PS C:\Users\exami\Desktop\git-assignment>
```

- Create new branch using **git checkout -b feature2**

```
PS C:\Users\exami\Desktop\git-assignment> git checkout -b feature2
Switched to a new branch 'feature2'
PS C:\Users\exami\Desktop\git-assignment> git log
commit ac1186fdb0f3adec3709f2dbfec89d7193eb086d (HEAD -> feature2, origin/main, origin/HEAD, main)
Author: Gourshab <ravindragourcode786@gmail.com>
Date:   Wed Mar 11 10:27:02 2026 +0530

    added index file

commit 858a0353cd99f903b2e8b7e9d9f263e119538911
Author: Ravindra Gour <143959353+gourshabrg@users.noreply.github.com>
Date:   Wed Mar 11 09:26:33 2026 +0530

    Initial commit
```

- Adding a new file **about.txt in feature2 branch.**

```
PS C:\Users\exami\Desktop\git-assignment> echo feature2 branch > about.txt
PS C:\Users\exami\Desktop\git-assignment> git add about.txt
PS C:\Users\exami\Desktop\git-assignment> git commit -m "Added about file"
[feature2 cf9ff24] Added about file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 about.txt
PS C:\Users\exami\Desktop\git-assignment> git push origin feature2
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 349 bytes | 349.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'feature2' on GitHub by visiting:
remote:      https://github.com/gourshabrg/git-assignment/pull/new/feature2
remote:
To https://github.com/gourshabrg/git-assignment
 * [new branch]      feature2 -> feature2
PS C:\Users\exami\Desktop\git-assignment>
```



7. Merge Branches

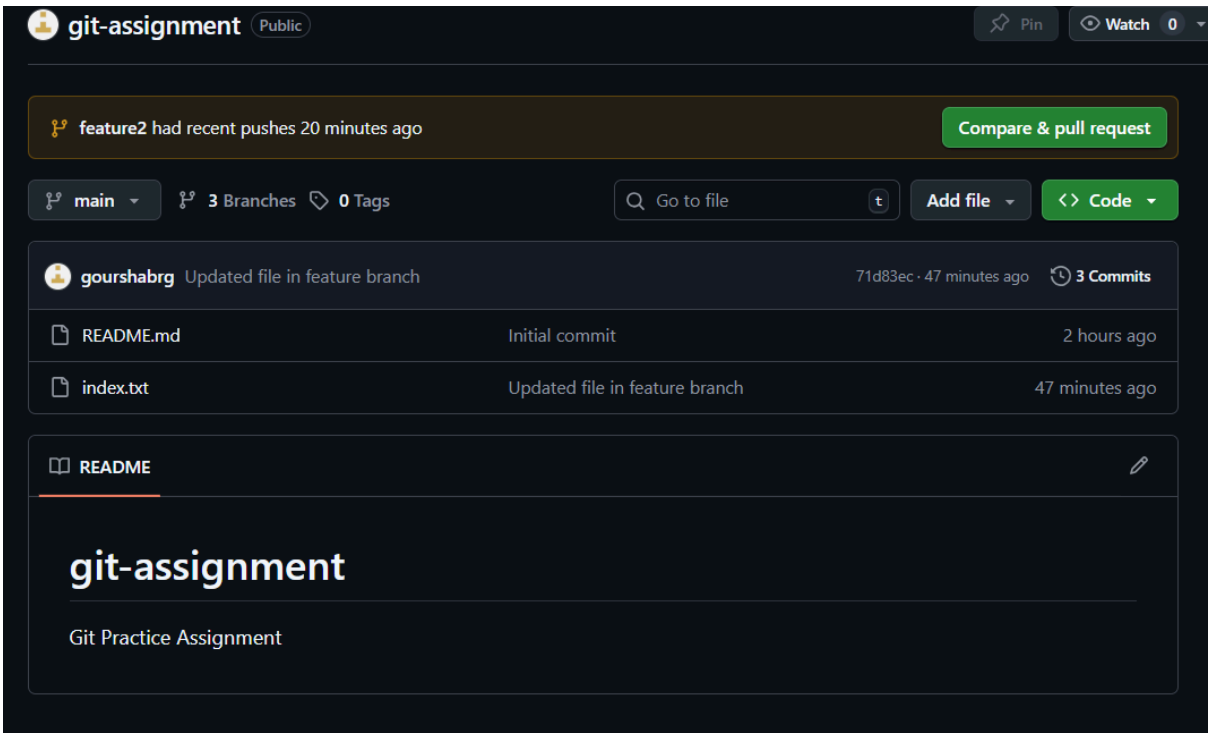
- **git checkout main**

  About: This command switches your working directory to the main branch.
- **git merge [name_of_branch]**

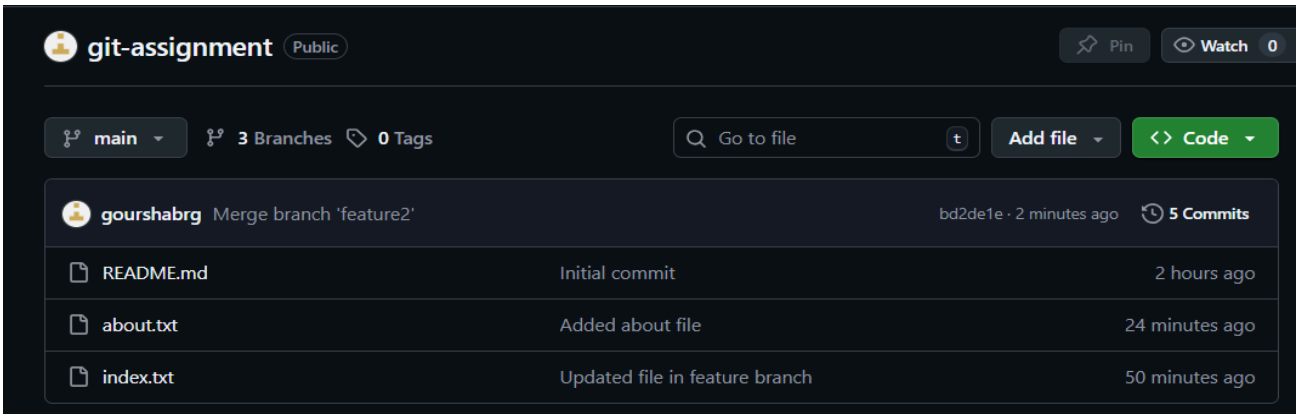
  About: This command combines the changes from feature1 branch into the current branch (main).
- **git push origin main**

  About: This command uploads the updated main branch to the remote repository (GitHub).

```
PS C:\Users\exami\Desktop\git-assignment> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\exami\Desktop\git-assignment> git merge feature1
Updating ac1186f..71d83ec
Fast-forward
 index.txt | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\exami\Desktop\git-assignment> git merge feature1
Already up to date.
PS C:\Users\exami\Desktop\git-assignment> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/gourshabrg/git-assignment
   ac1186f..71d83ec  main -> main
PS C:\Users\exami\Desktop\git-assignment>
```



- Same steps follow for the merging of the one second branch.



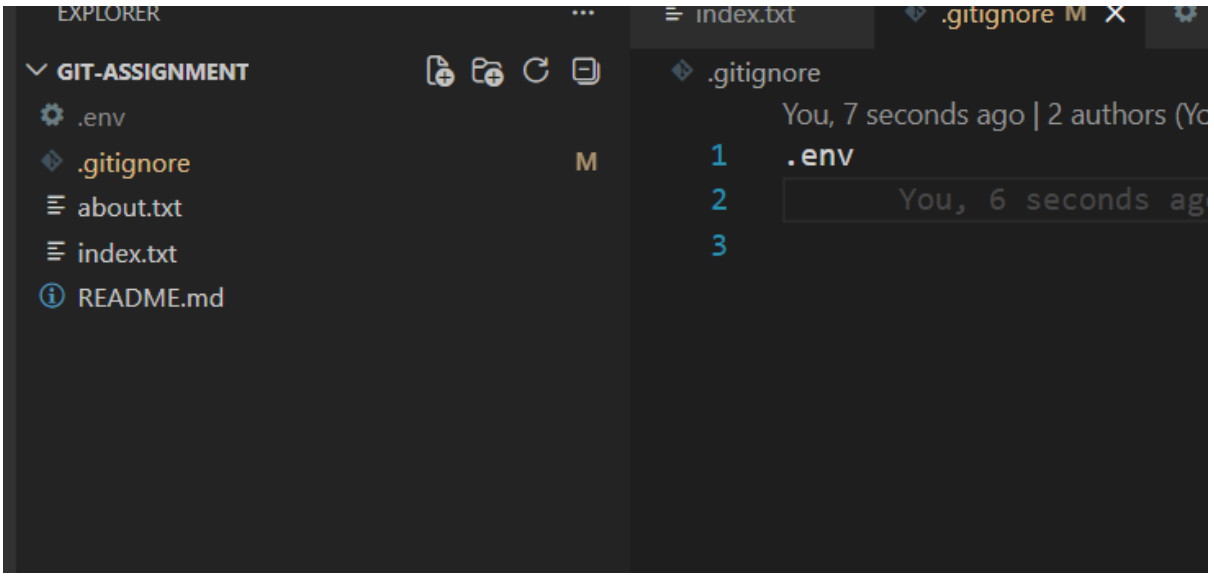
Summary: first we go to where we can merge using the **git checkout [branch_name].**

- Then which branch we want to merge in the current branch in local using **git merge [branch_name].**
- Then push all the changes cloud(Github)

.

8. Create an .env file, add the email to it, add that .env in .gitignore

- Create file **echo "EMAIL=ravindra@example.com" > .env**
- Also create the git ignore file and adding env file **echo ".env" > .gitignore**



9. Pull Changes from Remote

- Make changes in github repo file **text.txt** then we pull in local using **git pull origin main**

```
PS C:\Users\exami\Desktop\git-assignment> git pull origin main
From https://github.com/gourshabrg/git-assignment
 * branch            main       -> FETCH_HEAD
Updating ea6d7b6..7b37a06
Fast-forward
 index.txt | 1 +
 1 file changed, 1 insertion(+)
PS C:\Users\exami\Desktop\git-assignment>
```

-

10. View Git Log and History

- **git log**
  About: displays the commit history of a repository including **commit ID, author, date, and message**.

```
PS C:\Users\exami\Desktop\git-assignment> git log
commit 7b37a0679211edf0702eda11987b1ceac8e5ef18 (HEAD -> main, origin/main, origin/HEAD)
Author: Ravindra Gour <143959353+gourshabrg@users.noreply.github.com>
Date:   Wed Mar 11 12:17:13 2026 +0530

    Update index.txt with pull request note

    Add note about making changes for pull request
```

- **git log --oneline**
  About: A shortened version of commit history where **each commit is displayed in a single line**.

```
7b37a06 (HEAD -> main, origin/main, origin/HEAD) Update index.txt with pull request note
ea6d7b6 Add .env to .gitignore
cb38761 Added gitignore for env
bd2de1e Merge branch 'feature2'
cf9ff24 (origin/feature2, feature2) Added about file
7b37a06 (HEAD -> main, origin/main, origin/HEAD) Update index.txt with pull request note
ea6d7b6 Add .env to .gitignore
cb38761 Added gitignore for env
bd2de1e Merge branch 'feature2'
cf9ff24 (origin/feature2, feature2) Added about file
71d83ec (origin/feature1, feature1) Updated file in feature branch
ac1186f added index file
858a035 Initial commit
~
```

11. Git Status and Staging

- Check file status: **git status**

```
PS C:\Users\exami\Desktop\git-assignment> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   .gitignore

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\exami\Desktop\git-assignment>
```

- Stage files: **git add .**

```
PS C:\Users\exami\Desktop\git-assignment> git add .
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\exami\Desktop\git-assignment> git commit -m "staging demo"
[main f3ce9ce] staging demo
 1 file changed, 0 insertions(+), 0 deletions(-)
PS C:\Users\exami\Desktop\git-assignment> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 351 bytes | 351.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/gourshabrg/git-assignment
   7b37a06..f3ce9ce  main -> main
PS C:\Users\exami\Desktop\git-assignment>
```

12. Revert and Reset

- **git revert <commit_id>**

  About: undo a commit by creating a new commit that reverses the changes.
  It **does NOT delete history**.

Ex :

Initial commit history:

A --- B --- C --- D

If we run:

git revert D

Now history becomes:

A --- B --- C --- D --- E

```
PS C:\Users\exami\Desktop\git-assignment> echo demo revert > demo.txt
PS C:\Users\exami\Desktop\git-assignment> git add demo.txt
PS C:\Users\exami\Desktop\git-assignment> git commit -m "Add demo file for revert"
[main 09b7e30] Add demo file for revert
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 demo.txt
PS C:\Users\exami\Desktop\git-assignment> git log --oneline
09b7e30 (HEAD -> main) Add demo file for revert
f3ce9ce (origin/main, origin/HEAD) staging demo
7b37a06 Update index.txt with pull request note
ea6d7b6 Add .env to .gitignore
cb38761 Added gitignore for env
bd2de1e Merge branch 'feature2'
cf9ff24 (origin/feature2, feature2) Added about file
71d83ec (origin/feature1, feature1) Updated file in feature branch
ac1186f added index file
858a035 Initial commit
PS C:\Users\exami\Desktop\git-assignment> ^C
PS C:\Users\exami\Desktop\git-assignment> git revert 09b7e30
[main 8633206] Revert "Add demo file for revert"
 1 file changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 demo.txt
PS C:\Users\exami\Desktop\git-assignment> git log --oneline
8633206 (HEAD -> main) Revert "Add demo file for revert"
09b7e30 Add demo file for revert
f3ce9ce (origin/main, origin/HEAD) staging demo
7b37a06 Update index.txt with pull request note
ea6d7b6 Add .env to .gitignore
cb38761 Added gitignore for env
bd2de1e Merge branch 'feature2'
cf9ff24 (origin/feature2, feature2) Added about file
71d83ec (origin/feature1, feature1) Updated file in feature branch
ac1186f added index file
858a035 Initial commit
PS C:\Users\exami\Desktop\git-assignment>
```

- **git reset**
  About: moves the HEAD pointer to a previous commit and can remove commits and changes from the history.
  Ex:
  Initial commit history:
  A --- B --- C --- D
  If we run:
  git reset C

New history: becomes:
A --- B --- C
Types of Git Reset:

- **git reset --soft <commit_id>**
  About: Removes the last commit but **keeps all changes** staged
- **git reset <commit_id>**
  About: Removes the last commit and **unstages the changes**, but **keeps them in the working directory.**
- **git reset --hard <commit_id>**
  About: Removes the commit and **deletes all changes permanently** from staging and working directory.

```
PS C:\Users\exami\Desktop\git-assignment> git reset --soft 09b7e30
PS C:\Users\exami\Desktop\git-assignment> git log --oneline
09b7e30 (HEAD -> main) Add demo file for revert
f3ce9ce staging demo
7b37a06 Update index.txt with pull request note
ea6d7b6 Add .env to .gitignore
cb38761 Added gitignore for env
bd2de1e Merge branch 'feature2'
cf9ff24 (origin/feature2, feature2) Added about file
71d83ec (origin/feature1, feature1) Updated file in feature branch
ac1186f added index file
858a035 Initial commit
PS C:\Users\exami\Desktop\git-assignment>
```

More about : [Revert and Reset]